Module 1: Fundamentals of Software Testing and Quality Assurance

(Syllabus topics: Role of testing in software development, Importance of processes in software quality assurance, Faults, errors, and failures in software testing, Limitations and challenges in software testing, Concepts of verification and validation in software quality assurance.)

From CXT434-A.pdf (April 2025 Exam):

Part A (3 marks each):

1. What are software errors and faults. (3)

Part B (14 marks each):

11 a) Explain the difference between quality control and quality assurance. (7)

b) Explain the comparison between verification and validation (7)

OR

12 a) Explain in detail software quality and different views of software quality. (7)

b) What is testing? Explain the need of software testing in software development. (7)

From CXT434-B.pdf (September 2025 Exam):

Part A (3 marks each):

1. What are the main objectives of software testing? (3)
2. Define verification and validation in testing phase. (3)

Part B (14 marks each):

11 a) Define software testing. What is the role of testing in ensuring software quality? (7)

b) Differentiate between static analysis and dynamic analysis. (7)

OR

12 a) Explain the five views of software quality. How do these views contribute to understanding and managing software quality? (14)

From qp_stqa.pdf (Model Question Paper):

Part A (3 marks each):

1. Differentiate error, defect, and failure? (3)
2. Define verification and validation in testing phase. (3)

Part B (14 marks each):

11 (a) What is testing? Explain the need of software testing in software development. (7)

(b) Discuss the limitations and challenges in software testing. (7)

OR

12 (a) Explain the difference between quality control and quality assurance. (7)

(b) Discuss the applicability and relevance of life cycle models to verification and validation. (7)

Module 2: Testing Techniques and Methodologies

(Syllabus topics: White box and black box testing techniques – requirements-based testing, boundary value analysis, equivalence partitioning; White box testing – static analysis, unit testing, control flow testing; Integration, system, and acceptance testing methodologies; Non-functional testing techniques – performance, security, and usability testing.)

From CXT434-A.pdf (April 2025 Exam):

Part A (3 marks each):

3. What is the main goal of functional testing. (3)
4. Explain black box testing. (3)

Part B (14 marks each):

13 a) Describe the classification of white box testing. (7)

b) Explain the different techniques used in black box testing. (7)

OR

14 a) Explain in detail the different phases of software testing. (7)

b) Explain in detail non-functional testing. (7)

From CXT434-B.pdf (September 2025 Exam):

Part A (3 marks each):

3. What is the role of a "stub" and a "driver" in integration testing? (3)
4. Differentiate between aesthetic testing and accessibility testing. (3)

Part B (14 marks each):

13 a) Explain the different types of white box testing techniques and how can they be classified based on their focus and application in software testing? (8)

b) Define usability testing. Explain its importance in software development and user experience design. (6)

OR

14 a) Explain the importance of testing at boundary values in software systems. How do these tests help in uncovering defects that might not be found by random or general testing?
Consider an online voting system where the age must be between 18 and 120 years. Using boundary value analysis, create test cases to validate this input field. (14)

From qp_stqa.pdf (Model Question Paper):

Part A (3 marks each):

3. Discuss the challenges in white box testing. (3)
4. Compare walkthroughs and inspections. (3)

Part B (14 marks each):

13 (a) Explain the various methods to achieve static testing by humans. (8)
(b) Discuss the importance of traceability matrix. (6)

OR

14 (a) Describe the classification of white box testing. (7)
(b) Describe the various reasons for using black box testing. An input value for a product code in an inventory system is expected to be present in a product master table. Identify the set of equivalence classes to test these requirements. (7)

Module 3: Quality Assurance Principles and Standards

(Syllabus topics: Development of quality plans and objectives, Total Quality Management (TQM) concepts, Evaluation of quality models and standards (ISO, CMM, Six Sigma), Addressing quality challenges, Significance of national quality awards.)

From CXT434-A.pdf (April 2025 Exam):

Part A (3 marks each):

2. What are the software quality factors. (3)
5. Explain the concept of total quality management (TQM). (3)
6. List and explain the maturity levels of CMM. (3)

Part B (14 marks each):

15 a) Discuss what is six sigma quality. How is it achieved? (7)

15 a) Discuss what is six sigma quality. How is it achieved? (7)

b) Explain the relationship between quality factor and criteria. (7)

OR

16 a) List and explain the principles of ISO 9000:2000 standard. (7)

b) Explain in detail CMM (Capability Maturity Model). (7)

From CXT434-B.pdf (September 2025 Exam):

Part A (3 marks each):

5. What is quality standards? (3)
6. Explain the concept of six sigma and its significance in process improvement. (3)

Part B (14 marks each):

15 a) Sketch a diagram of the five maturity levels in the capability maturity model. Discuss the progression of an organization through these levels, highlighting the key characteristics and improvements associated with each level. (14)

OR

16 a) Explain the eight principles of the ISO 9000:2000 software quality standard. (8)

b) Discuss the relationship between quality factor and criteria. (6)

From qp_stqa.pdf (Model Question Paper):

Part A (3 marks each):

5. List and explain the maturity levels of CMM. (3)
6. What is total quality management (TQM)? (3)

Part B (14 marks each):

15 (a) Discuss what is six sigma quality. How is it achieved? (7)

(b) Describe the basic concepts of total quality management. (7)

OR

16 (a) List and explain the principles of ISO 9000:2000 standard. (7)

(b) Discuss the relationship between quality factor and criteria. (7)

Module 4: Regression Testing and Test Management

(Syllabus topics: Importance of regression testing in software maintenance, Regression test planning and case selection, Dynamic slicing and test minimization techniques, Tools for regression testing, Test planning, execution, and reporting.)

From CXT434-A.pdf (April 2025 Exam):

Part A (3 marks each):

7. Explain Test Reporting. (3)

8. Explain the process of regression testing. (3)

Part B (14 marks each):

17 a) Write a short note on tools for regression testing. (7)

b) What is program slicing? Explain dynamic program slicing. (7)

OR

18 a) Explain in detail test planning. (7)

b) Explain the Importance of regression testing in software maintenance. (7)

From CXT434-B.pdf (September 2025 Exam):

Part A (3 marks each):

7. Explain the process of dynamic slicing. (3)

8. What is a test plan? (3)

Part B (14 marks each):

17 a) Explain the role of selenium in regression testing. Discuss its features, advantages and how it can be used to automate regression test cases. (7)

b) Write a short note on tools for regression testing. (7)

OR

18 a) Discuss the role of test reporting in regression testing and explain how different types of test reports help stakeholders assess testing progress and software quality. (14)

From qp_stqa.pdf (Model Question Paper):

Part A (3 marks each):

7. Discuss the role of regression testing when developing a new software release. (3)

8. Discuss the importance of Test plan. (3)

Part B (14 marks each):

17 (a). When regression testing is done ? Discuss the difference in regression testing for a major release versus minor release of a product. (7)

(b). Write a short note on tools for regression testing. (7)

OR

18 a. What is program slicing? Explain dynamic program slicing. (6)
b. Describe the techniques of executing tests and reporting results. (8)

Module 5: Software Test Automation and Object-Oriented Testing
(Syllabus topics: Scope and benefits of test automation, Design and implementation of automation frameworks, Utilization of testing tools for automation, Object-oriented testing principles, Analysis of software quality metrics and continuous improvement initiatives.)

From CXT434-A.pdf (April 2025 Exam):
Part A (3 marks each):
9. What is the scope of automation testing? (3)
10. Define software quality metric. (3)

Part B (14 marks each):
19 a) Discuss the criteria and steps in selecting the testing tool for automation. (7)
b) Explain the framework for test automation. (7)

OR

20 a) Explain design and architecture for automation. (7)
b) Explain the continuous improvement initiatives in software quality. (7)

From CXT434-B.pdf (September 2025 Exam):
Part A (3 marks each):
9. Explain the key benefits and challenges of test automation. (3)
10. Explain the principles of object-oriented testing. (3)

Part B (14 marks each):
19 a) Draw a diagram representing the key components of test automation. Explain the role of each component in the automation framework and how they work together to ensure efficient test execution. (8)
b) Explain the different types of software quality metrics. (6)

OR

20 a) Describe the characteristics of conventional algorithm-centric programming languages and explain how they focus on algorithm development in software programming. (14)

From qp_stqa.pdf (Model Question Paper):

Part A (3 marks each):

9. Define encapsulation and polymorphism. (3)

10. Define quality metric. (3)

Part B (14 marks each):

19 a. Explain the framework for test automation. (8)

b. Discuss the criteria and steps in selecting the testing tool for automation. (6)

OR

20 a. Describe the tools for testing of object oriented systems. (6)

b. Why do integration and system testing assume special importance for an object oriented system? (8)